

CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING

Abstract

Credit card fraud is one of the major challenges faced by financial institutions due to the increasing use of online transactions. The objective of this project is to develop a Machine Learning model capable of identifying fraudulent credit card transactions. The dataset was preprocessed, normalized, and balanced using SMOTE to handle class imbalance. Logistic Regression and Random Forest algorithms were trained and evaluated using Precision, Recall, and F1-Score. The Random Forest model achieved the best performance and successfully detected fraudulent transactions with high accuracy.

1. Introduction

With the rapid growth of digital payments, credit card fraud has become a significant concern for banks and customers. Fraudulent transactions can result in financial losses and security risks. Machine Learning techniques can help identify suspicious transactions automatically by analyzing transaction patterns and classifying them as genuine or fraudulent.

This project focuses on building a fraud detection system using supervised Machine Learning algorithms.

2. Objective

The objectives of this project are:

- To build a Machine Learning model for fraud detection.
- To preprocess and normalize transaction data.
- To handle class imbalance using SMOTE.
- To train classification models.
- To evaluate model performance using Precision, Recall, and F1-Score.

3. Dataset Description

The dataset contains credit card transactions performed by European cardholders.

Features

- Time: Time elapsed between transactions.
- V1 to V28: Principal components obtained through PCA transformation.
- Amount: Transaction amount.
- Class:
 - 0 = Genuine Transaction
 - 1 = Fraudulent Transaction

The dataset is highly imbalanced, with fraudulent transactions representing a very small percentage of total transactions.

4. Methodology

4.1 Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked class distribution.
- Normalized the Amount feature using StandardScaler.
- Separated features and target variables.

4.2 Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

4.3 Handling Class Imbalance

Since fraud cases were significantly fewer than genuine transactions, SMOTE (Synthetic Minority Oversampling Technique) was applied to generate synthetic fraud samples and balance the dataset.

4.4 Model Training

Two Machine Learning models were trained:

1. Logistic Regression
 2. Random Forest Classifier
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5. Evaluation Metrics

The models were evaluated using:

Precision

Measures the proportion of correctly predicted fraud transactions among all transactions predicted as fraud.

Recall

Measures the proportion of actual fraud transactions correctly identified by the model.

F1-Score

Harmonic mean of Precision and Recall.

Confusion Matrix

Provides detailed information about correct and incorrect predictions.

6. Results

Logistic Regression

Confusion Matrix:

Actual/Predicted	Genuine	Fraud
Genuine	56162	702
Fraud	10	88

Performance:

- Precision: 0.11
- Recall: 0.90
- F1-Score: 0.20

Although Logistic Regression achieved high recall, it produced many false positives.

Random Forest Classifier

Performance:

Metric	Score
Precision	0.83
Recall	0.83
F1-Score	0.83

The Random Forest model significantly improved fraud detection performance and reduced false predictions.

7. Conclusion

A Credit Card Fraud Detection system was successfully developed using Machine Learning techniques. Data preprocessing and SMOTE were used to handle class imbalance effectively. Both Logistic Regression and Random Forest models were evaluated. Random Forest demonstrated superior performance with a Precision, Recall, and F1-Score of 0.83 for fraud detection.

The project highlights the importance of Machine Learning in detecting fraudulent financial transactions and improving transaction security.

8. Future Scope

- Use advanced ensemble techniques such as XGBoost and LightGBM.
 - Implement real-time fraud detection systems.
 - Experiment with Deep Learning models.
 - Improve performance using feature engineering and hyperparameter tuning.
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Tools and Technologies Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Scikit-learn
 - Imbalanced-learn (SMOTE)
 - Machine Learning
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References

1. Kaggle Credit Card Fraud Detection Dataset.
2. Scikit-learn Documentation.
3. Imbalanced-learn Documentation.
4. Machine Learning Research Papers on Fraud Detection.